

Nickname:

1.4 Classwork: Standard Deviation

Use your GDC. Give answers exactly, or correct to three significant figures. Write down the values you put into the calculator.

1. A supermarket records the number of items bought by each of the 50 customers who used the express checkout one morning.

Number of items, x	1	2	3	4	5	6
Frequency, f	4	9	13	11	8	5

- (a) Find the mean number of items bought.
- (b) Find the standard deviation.
- (c) Write down the median, the lower quartile Q_1 and the upper quartile Q_3 , and hence find the interquartile range.
- (d) Find the number of customers who bought more than one standard deviation above the mean.
- (e) A fifty-first customer goes through the express checkout with 20 items. State the effect this has on the mean and on the standard deviation. Give a reason for your answer.

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2. Two machines fill bottles that are labelled 500 millilitres. Eight bottles are taken from each machine and the volume of each, in millilitres, is measured.

Machine A: 498, 502, 497, 503, 500, 499, 501, 500

Machine B: 494, 507, 499, 505, 496, 503, 497, 499

- (a) Find the mean and the standard deviation of the volumes from Machine A.
- (b) Find the mean and the standard deviation of the volumes from Machine B.
- (c) The manufacturer wants every bottle to be as close to 500 ml as possible. State which machine you would choose, and justify your answer by referring to both statistics.
- (d) Machine A is adjusted so that every bottle it fills contains 3 ml more than before. Write down the new mean and the new standard deviation for Machine A.
- (e) Find the number of the eight Machine B bottles whose volume lies within one standard deviation of the mean.

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